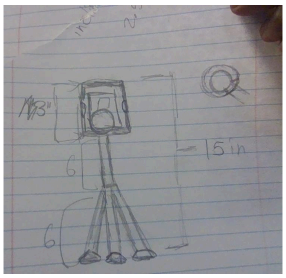
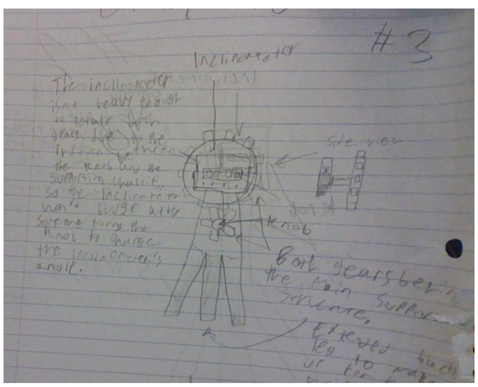
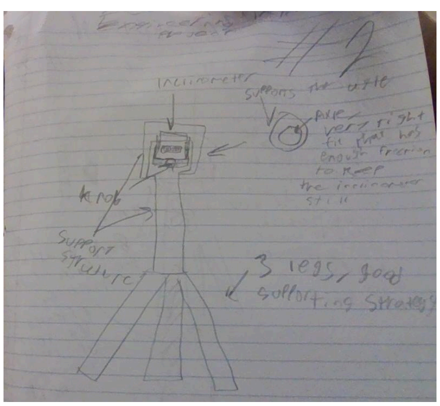
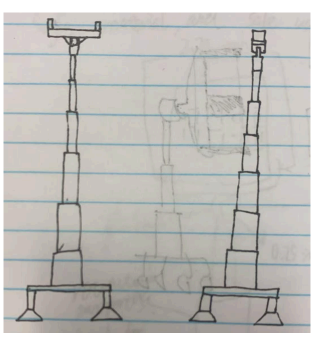
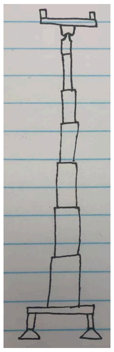
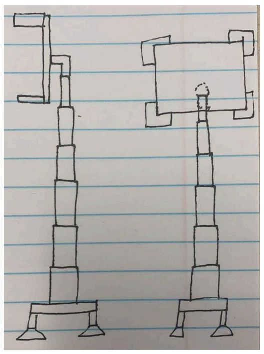
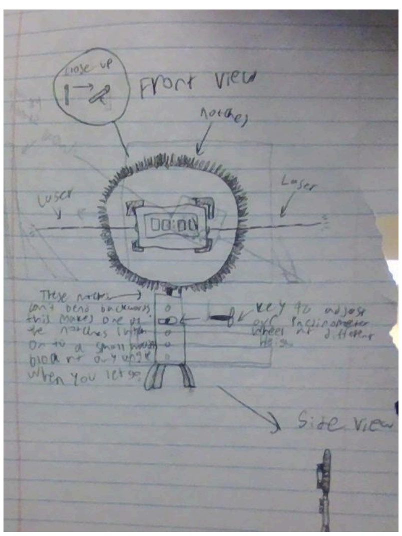
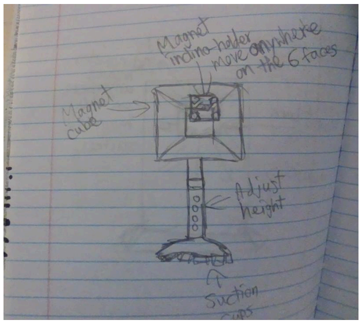
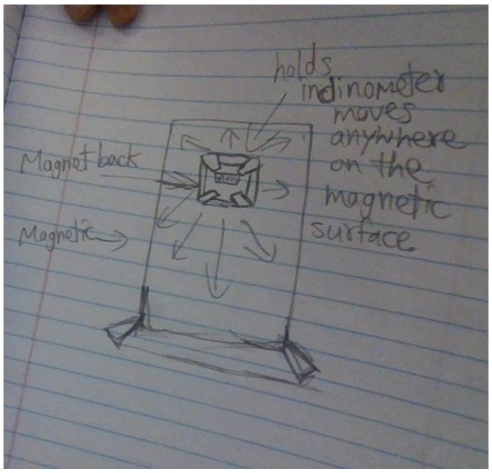
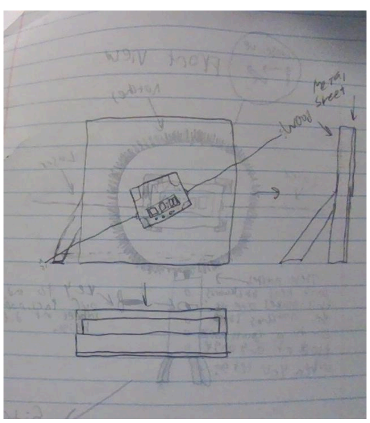
D1—Documentation of our ideas

<table>
<tr><td>Solution #1: </td><td>Date added: 04/22/2026</td></tr>
<tr><td>Solution #2: 

Classic tripod mechanism. An inclinometer holder that rotates with a ball and socket to rotate the inclinometer. The holder's height can go down the neck. Everything is 3D printed. Suction cup legs.</td><td>Date added: 04/22/2026</td></tr>
<tr><td>Solution #3: 
</td><td>Date added: 04/22/2026</td></tr>
<tr><td>Solution #4: 
</td><td>Date added: 04/22/2026</td></tr>
</table>

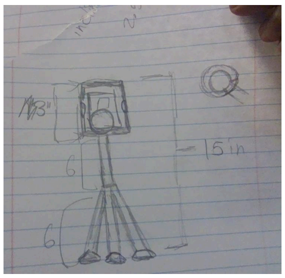
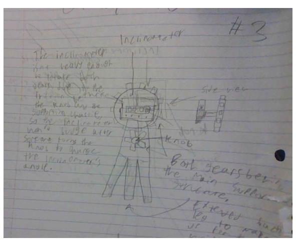
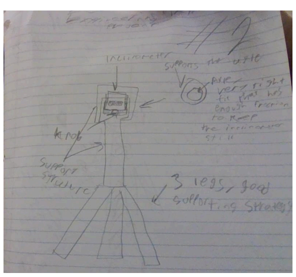
Solution #5:  Inclinometer holder that rotates on an axis. Height can be adjusted. Everything is made with a 3D printer. Rubber-covered stand legs.	Date added: 04/23/2026
Solution #6: 	Date added: 04/23/2026
Solution #7: 	Date added: 04/24/2026
Solution #8:  1 mm thick for the flat base; 0.5” height for each leg of the base; 3” each section for the retractable stick; 180° adjustable platform on each side, left and right.	Date added: 04/25/2026

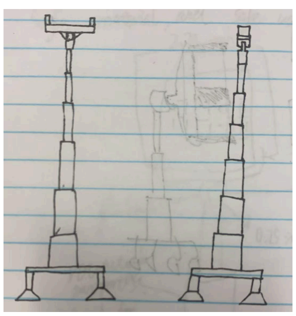
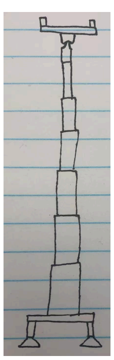
Solution #9:  Only the platform setting is different from Solution #11: The platform can rotate by the stick as the axle 360°.	Date added: 04/25/2026
Solution #10:  Only the platform setting is different from Solution #11: The platform can rotate 360° on each side, left and right; it has four fences on each of its corners, 1” high.	Date added: 04/25/2026
Solution #11:  The inclinometer is attached to a “case” which is glued onto a wheel. All of the surface area on the circumference of the wheel is covered with little switches that can only bend one way. When this wheel is attached to an axle, it can rotate and stop at any point, since a node will bend back against a small block and stop the wheel from moving. This block is located in the middle of the inclinometer stand. This design also features a collapsing hole/key mechanism to adjust height at will.	Date added: 04/25/2026

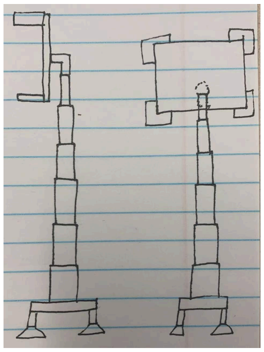
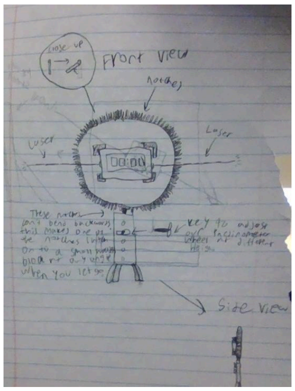
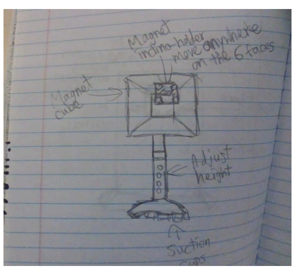
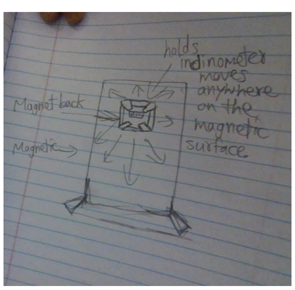
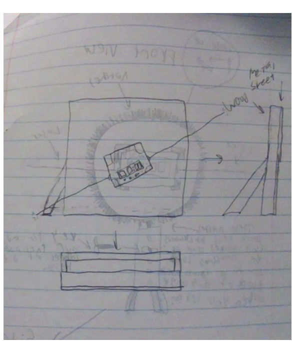
Solution #12:  Magnetic cube on top with adjustable height. Suction cup on the bottom of the legs.	Date added: 04/27/2026
Solution #13:  Fully magnetic metal plate. The bottom stand is made of a soft rubber material to balance it. The inclinometer holder has a metal magnetic back to move freely on the surface.	Date added: 04/27/2026
Solution #14:  A slab of wood with a thin metal plate layered on the end, then a small case that can hold the inclinometer from all four sides magnetically, and has a magnet on the back that can attach to the metal plate. Wooden or 3d printed supports are on the back of the slab that keep it upright.	Date added: 04/27/2026

D2—Design Solution Choice

Idea/Concept for the project	Unique details about this idea	Build it? Yes/no	Date

	A tripod mechanism, inclinometer is contained in a holder where it rotates horizontally. The magnetic back will be used to rotate the holder. The legs will contain rubber caps to prevent any movement from the floor.	No	04/22/2026
	Classic tripod mechanism. An inclinometer holder that rotates with a ball and socket to rotate the inclinometer. The holder's height can go down the neck. Everything is 3D printed. Suction cup legs.	No	04/22/2026
	This inclinometer stand uses 3 supports at the bottom and a linear structure in the middle. The inclinometer is attached to a “case” which is glued to a a “gearbox” style design to maintain friction, retain an angle, and keep it stationary.	No	04/22/2026
	3 legs supporting the device on the bottom, then a linear, straight pole for the middle section, then a knob that uses an axle that is attached to the bottom of the inclinometer, which uses friction to keep the angle still.	No	04/22/2026
	Holder rotates on one axis; adjustable height; fully 3D printed; rubber-covered stand legs.	No	04/23/2026

	A holder with magnetic corners that holds a magnetic inclinometer. Adjustable height with a tightening mechanism. Rubber-tipped stand bottom legs.	No	04/23/2026
	The holder that rotates on a cylinder that is on the back. Adjustable height and suction cups on bottom of stand legs for stability.	No	04/24/2026
	Flat base with 1 mm thickness; 0.5 in leg height; 3 in retractable sections; platform adjusts 180° on left and right sides.	No	04/25/2026
	Same as Solution 11, but the platform rotates 360° using the stick as the axle.	No	04/25/2026

	Same base and retractable structure as Solution 11, but the platform rotates 360° on each side and has four 1 in fences for protection.	No	04/25/2026
	The inclinometer is attached to a “case” which is glued onto a wheel. All of the surface area on the circumference of the wheel is covered with little switches that can only bend one way. When this wheel is attached to an axle, it can rotate and stop at any point, since a node will bend back against a small block and stop the wheel from moving. This block is located in the middle of the inclinometer stand. This design also features a collapsing hole/key mechanism to adjust height at will.	No	04/25/2026
	Magnetic cube on top with adjustable height; suction cup on the bottom of the legs.	No	04/27/2026
	Fully magnetic metal plate; soft rubber bottom stand; inclinometer holder with magnetic back that can move freely.	No	04/27/2026
	A slab of wood with a thin metal plate layered on the end, then a small case that can hold the inclinometer from all four sides magnetically, and has a magnet on the back that can attach to the metal plate. Wooden or 3d printed supports are on the back of the slab that keep it upright.	Yes	04/27/2026

Requirement	Weight	1	2	3	4	5	6	7
1	5	4	4	3	4	5	4	4
2	5	3	3	3	3	4	4	4
3	5	4	4	4	4	4	4	4
4	3	3	3	4	4	4	4	4
5	3	3	5	2	3	4	4	4
6	3	4	4	4	4	4	4	5
7	3	3	3	4	4	4	4	4
8	3	4	4	2	4	5	4	4
9	3	4	4	3	4	5	4	4
10	1	4	4	4	4	4	4	4
11	1	5	5	5	5	5	5	5
Total		127	133	116	133	152	141	144

Requirement	Weight	8	9	10	11	12	13	14
1	5	4	3	3	2	4	4	5
2	5	4	3	5	2	4	4	5
3	5	5	5	5	4	4	5	4
4	3	4	4	5	5	3	2	5
5	3	4	5	5	5	4	2	3
6	3	5	4	5	3	4	4	5
7	3	4	3	3	2	3	4	5
8	3	4	4	3	2	4	5	4
9	3	4	4	4	3	4	5	3
10	1	4	4	4	4	4	4	5
11	1	5	5	5	5	5	5	5

Total		149	136	149	109	135	140	155

After comparing all 14 design ideas with our approved requirements, we decided to choose Solution 14. It got the highest total score in our design matrix. We think it is the best choice because it is safe, stable, strong, and simple enough to build within 10 days. The wood base, metal plate, magnetic case, and back supports should help keep the inclinometer in place without tipping over. Some other designs had more moving parts or more adjustment options, but Solution 14 seemed more practical and easier for us to actually build and use. That is why we chose Solution 14 as our final design.

D3—Our plan

Task	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Finalize dimensions and design details	■									
Create CAD parts in Onshape	■	■								
Check laser/display clearance and angle range		■								
Gather materials and prepare print files			■	■						
3D print parts				■						
Assemble prototype					■	■	■			
Safety and function testing							■	■		
Revise weak parts/reprint if needed								■	■	
Final testing									■	■
Add logo/design									■	■

D4—The blueprint

Material List						
Item Name	**Quantity**	**Length**	**Width**	**Height/Thickness**	**Cost**	**Link**

5/4 in. x 6 in. x 12 ft. Standard Ground Contact Pressure-Treated Lumber	1	16”	6”	5/4 in	9.99	LINK
430 Stainless Steel Metal Plates, 12" x 6" Inch, Suitable for Magnetic Installation Plate Walls, Suitable for Metal Plates in processes, Kitchens, and Offices (2, 0.5mm Thick)	1 order, 2 magnetic sheets	12”	6”	0.5 mm	7.99	LINK
30Mil Small Magnetic Sheet with Adhesive Backing - Magnet Stickers for Classroom Decorations- Teacher Must Have - Teacher Supplies, Small Index Card 2.5x3.5 for Pictures,28 Pack *Updated:* Magnets - 1,8 cm (0.7") Round Disc Ceramic Magnets - Flat Circle Magnets for Science & DIY - Ferrite Small Magnets Perfect for Refrigerator, Whiteboard, Fridge - 30 PCs	1 order. 14 magnet sheets out of the 28 magnet sheets. 8 magnets.	2.5in each given	3.5in each given	30 Mil given	9.99 9.99	LINK LINK